EMERGENCY DISPATCH RESILIENCE & FAILURE-HANDLING PLAN

A practical design note for safer ambulance allocation

Project: Ambulance and Emergency Dispatch System

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Why this document is different

A dispatch system should not only select an ambulance when everything works; it should also respond predictably when a resource is unavailable, an input is invalid, or an assignment conflicts with another request. This plan describes failure cases worth handling and testing in the project.

1. Failure scenarios and recommended system behaviour

Scenario	Expected safe behaviour	Validation idea
No ambulance available	Show a clear “no available ambulance” result; do not assign a busy unit.	Mark all ambulances busy and submit a request.
Duplicate assignment attempt	Reject the second assignment and preserve the ambulance’s existing status.	Send two requests for the same ambulance.
Invalid distance or route data	Reject invalid values and report that route calculation could not be completed.	Try a negative distance or disconnected route.
No hospital beds available	Check another suitable hospital; if none qualifies, report that no suitable option was found.	Set candidate hospitals to zero available beds.
Driver rejects assignment	Release the rejected assignment safely and evaluate the next eligible ambulance.	Simulate a rejection and verify reassignment.
Critical and routine requests arrive together	Process the critical request first according to the priority rule.	Queue requests with different severity levels.

2. Dispatch safeguards

• **Check before assigning:** confirm the ambulance is available immediately before changing its status to assigned.
• **Keep status consistent:** update the assignment record and ambulance status together in one controlled operation.
• **Validate inputs:** check emergency severity, ambulance availability, route distances and hospital bed counts.
• **Explain outcomes:** return a clear result for successful assignment, no resource available, invalid input or route failure.

3. Suggested validation checklist

Use this checklist during testing; record the actual result and mark Pass/Fail only after running each test.

Test	Expected result	Actual status
All ambulances busy	No assignment is made.	To be tested
Duplicate assignment	Second assignment is blocked.	To be tested
No hospital beds	Alternative hospital is checked.	To be tested
Invalid route data	Invalid input is rejected safely.	To be tested
Driver rejection	Next eligible ambulance is considered.	To be tested

4. Implementation note

This document is a proposed design and test plan, not evidence that every safeguard is already implemented. The Phase-II report describes a prototype with emergency prioritization, Dijkstra-based route calculation, ambulance assignment, duplicate-assignment blocking and hospital selection using sample data. The remaining cases above should be tested against the actual code before being reported as completed.

Suggested GitHub commit message

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Add dispatch failure-handling and validation plan
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Recommended filename: Emergency_Dispatch_Resilience_and_Failure_Handling.pdf